



NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science and Engineering
Examination	Mid Semester – 2024-25
Program	B. Tech / M. Tech / Ph. D
Session	Spring
Course code & Course Name	CS6404 - Software Testing
Marks	30
Duration	2 Hours

This question paper contains 2 pages

Instructions:

Answer all questions.

Q. No.	Questions	Marks	CO*	BL**
1.	Apply black-box testing on a function named find-rectangle-intersection that accepts four floating point numbers representing two pairs of co-ordinates (x1, y1), (x2, y2), (x3, y3), (x4, y4) and generate the test cases. The first two points (x1, y1) and (x2, y2) represent the lower left and upper right points of the first rectangle. The second two points (x3, y3) and (x4, y4) represent the lower left and upper right points of the second rectangle. It is assumed that the length and width of the rectangle are parallel to either the x-axis or y-axis. The function computes the points of intersection of the two rectangles and prints their points of intersection.	5	CO-1	BL-3
2.	Consider a system having an FSM for a gaming software named Gumball with the following states and transitions: States: NoQuarter, HasQuarter, Sold, SoldOut, Error Transitions: NoQuarter to HasQuarter: addQuarter() NoQuarter to NoQuarter: turnCrank() HasQuarter to Sold: turnCrank() Sold to SoldOut: takeGumball() SoldOut to Error: turnCrank(), addQuarter() HasQuarter to Error: addQuarter() Sold to Sold: turnCrank() Sold to NoQuarter: takeGumball() Design test cases for the above FSM using state table based testing.	5	CO-1	BL-4
3.	Consider the operation of the following washing machine: <i>The washing machine waits for the “Start” switch to be pressed. After the user presses the “Start” switch the machine fills the wash tub with either hot or cold water depending upon the setting of the “Hotwash” switch. The water filling continues until the high level is sensed. The machine starts the agitation motor and continues agitating the wash tub until either the present timer expires or the user presses the “Stop” switch. After the agitation stops, the machine waits for the user to press the “Start Drying” switch. After the user presses the “Start Drying” switch the machine starts the hot air blower and continues blowing hot air into the drying chamber until either the user presses the “Stop” switch or the present timer expires.</i> Design test cases for the above washing machine using decision table based testing.	5	CO-1	BL-4

4.	Consider the following C function named <i>sort</i>. <pre> /* sort takes an integer array and stores it in ascending order */ void sort (int a[], int n){ int i,j; for(i=0;i<n-1;i++) for(j=i+1;j<n;j++) if(a[i]>a[j]) { temp=a[i]; a[i]=a[j]; a[j]=temp; } } </pre> Draw the CFG for the above function. Determine the cyclomatic complexity. List all independent paths. Design test cases to cover all independent paths.	5	CO-2	BL-3
5.	Consider the program given in Q 4. Generate four mutants and design test cases taking into account each mutant, using mutation testing. Calculate mutation score of your test suite. Can you improve further the obtained mutation score?	5	CO-2	BL-3
6.	Design a test suite using MC/DC testing for the following expression: $a \ \&\& \ (b \ \ c)$. Why MC/DC testing is preferred in software industries?	5	CO-2	BL-3

CO**: Course Outcome; *BL**: Bloom's Taxonomy Level.